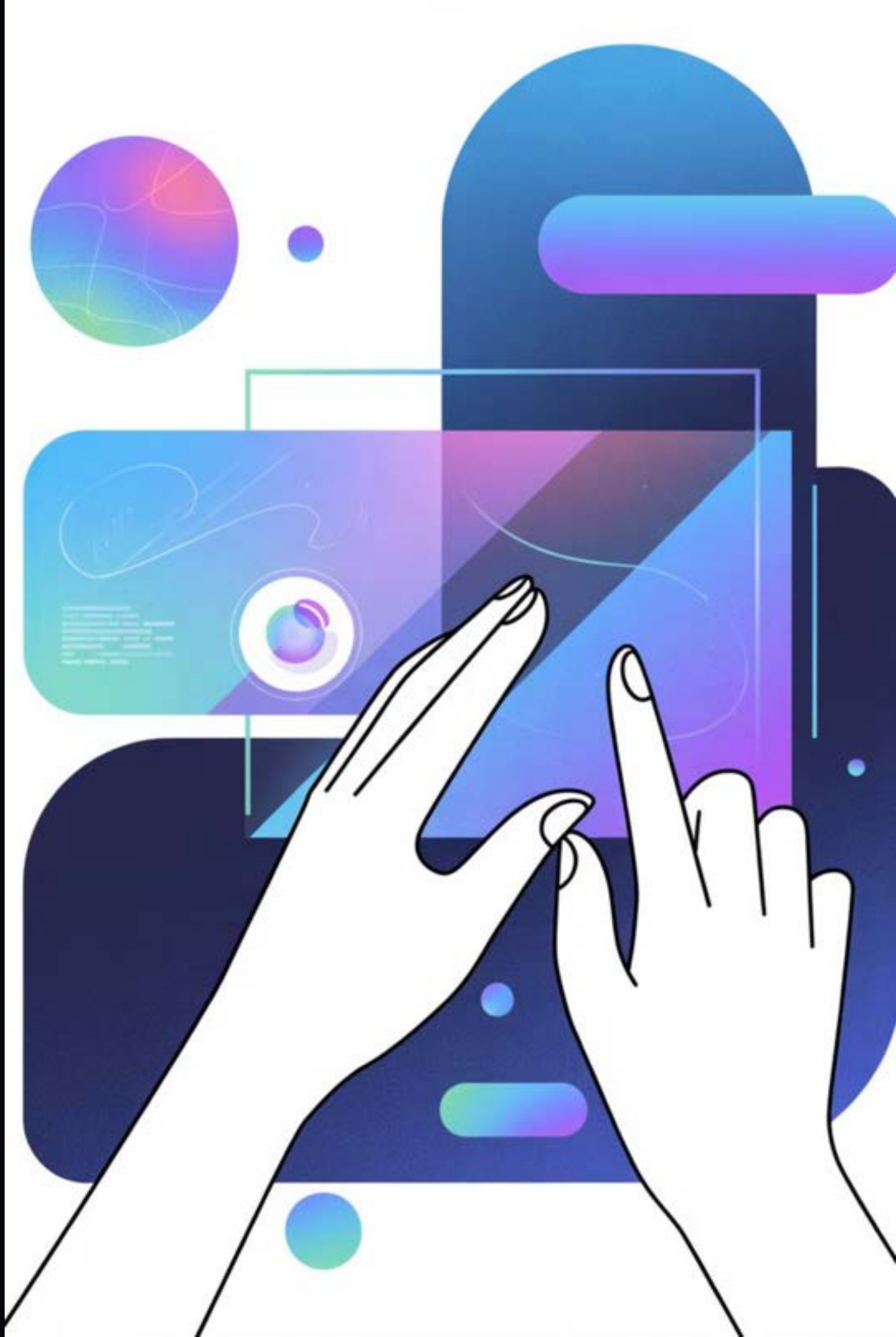
Real-time Explainable Gesture Recognition

Achieving near-perfect accuracy with transparent AI for human-computer interaction.



System Architecture: A Three-Pillar Approach

Preprocessing Pipeline

Prepare images and augment data

CNN Model

Extract features and classify gestures

Explainability Layer

Provide interpretable model insights

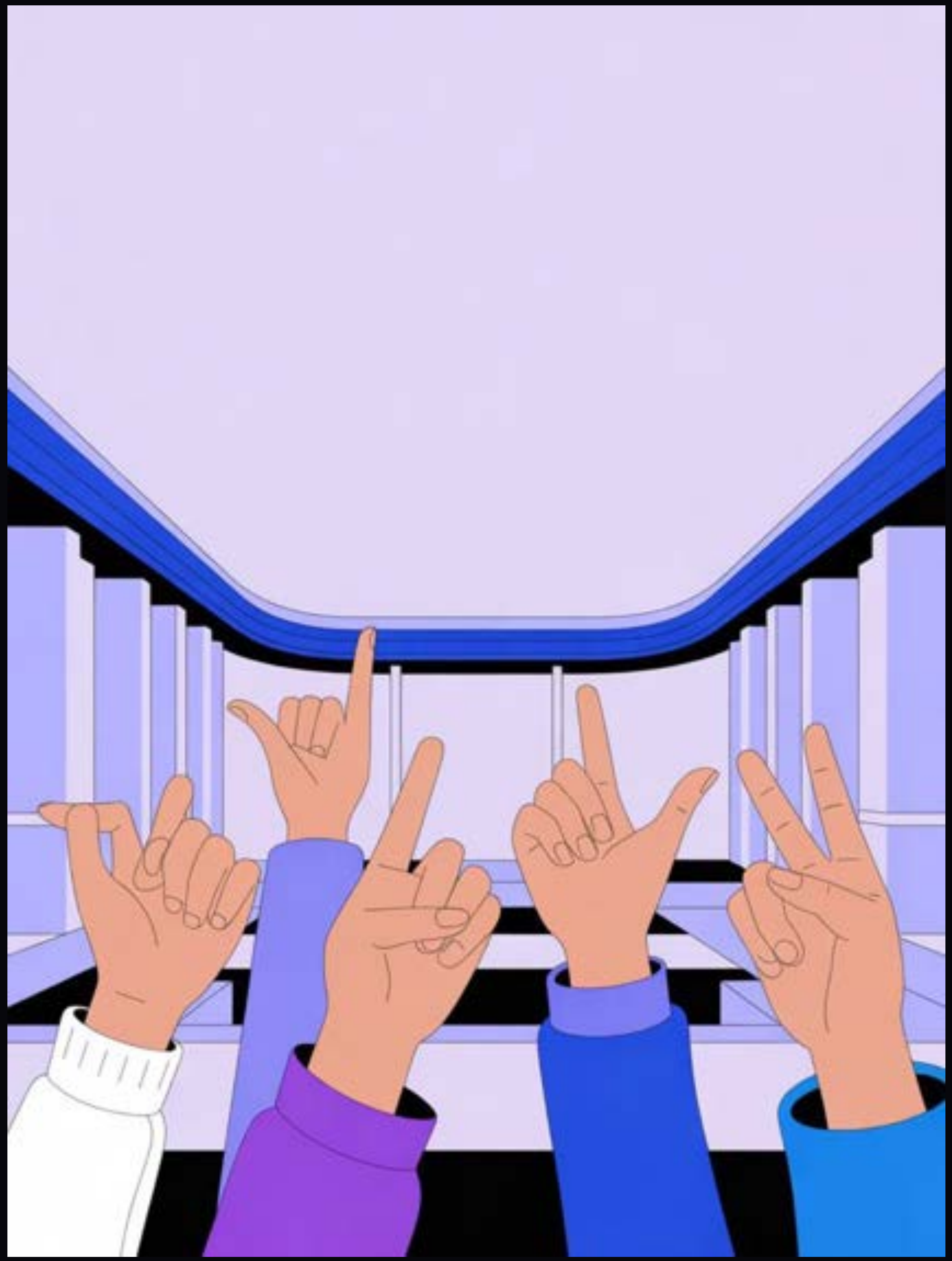
Our system integrates three core components to deliver robust and interpretable gesture recognition, from raw image input to confident classification.

DATASET

Leap Gesture Recognition Dataset

Selected for its clean image quality, the dataset comprises approximately **20,000 images** across 10 distinct gesture classes from 10 subjects. This rich data source was crucial for training a highly accurate model.

- 10 Gesture Classes: Palm, L, Fist, Fist_moved, Thumb, Index, Ok, Palm_moved, C, Down
- Dataset Split: 80% training, 20% validation
- Validation Samples: 3,200 with balanced class distribution (320 per gesture)

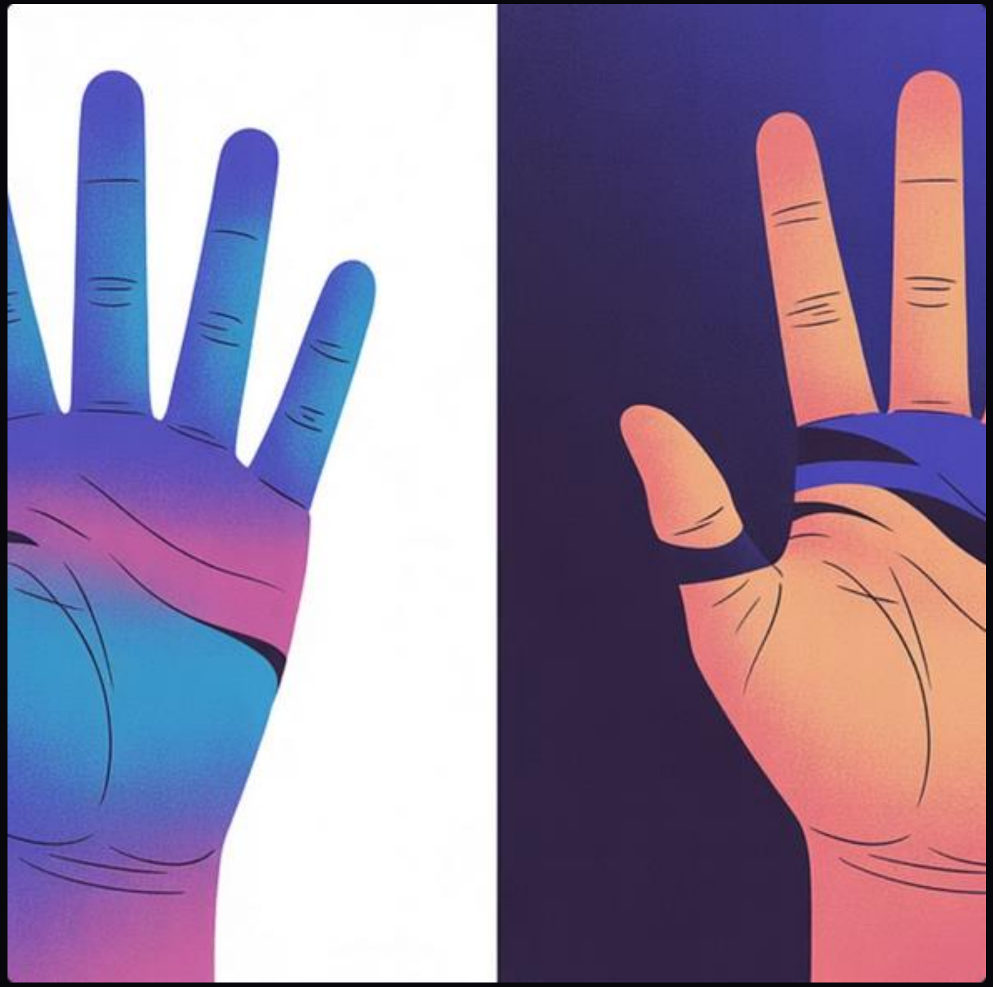


PREPROCESSING

Robust Image Enhancement & Segmentation

CLAHE in LAB Color Space

Enhances contrast while preventing noise amplification, ensuring clear hand features.



Hand Segmentation

Combines HSV skin color detection and adaptive thresholding for robust isolation from background elements.

Careful preprocessing ensures optimal input for the CNN, minimizing irrelevant data and emphasizing key features.

DATA AUGMENTATION

Expanding Dataset Diversity

To improve model generalization and robustness, we applied a suite of data augmentation strategies, simulating various hand positions and orientations without additional data collection.

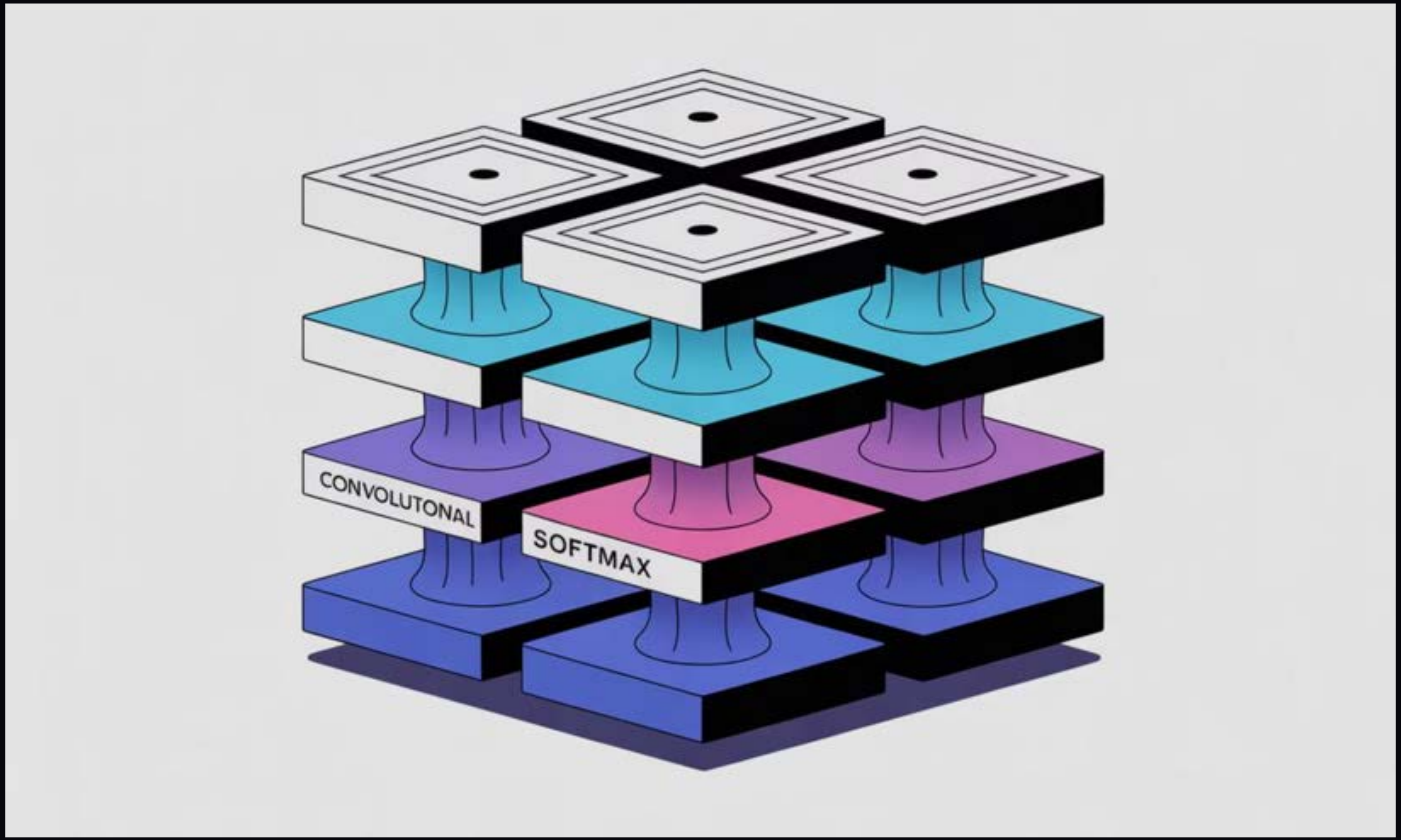
- Rotation: **±20°**
- Translation: **±20%**
- Shear Transformation: **±15%**
- Zoom Variation: **±20%**
- Horizontal Flipping

MODEL ARCHITECTURE

Deep Convolutional Neural Network (CNN)

Our classification model employs a deep CNN with **four convolutional blocks**, each featuring dual convolution layers with progressively increasing filters (32, 64, 128, 256). Batch normalization, max pooling, and dropout (0.25-0.5) prevent overfitting.

- Total Parameters: Approximately **2.5 million**
- Fully Connected Layers: 512 and 256 neurons with batch normalization and dropout
- Final Layer: Softmax output for 10-class classification



TRAINING & OPTIMIZATION

Efficient Training on GPU Infrastructure

Training utilized the Adam optimizer with an initial learning rate of 0.001 and categorical cross-entropy loss. A batch size of 64 optimized GPU memory on Kaggle's Tesla T4 x2 infrastructure.

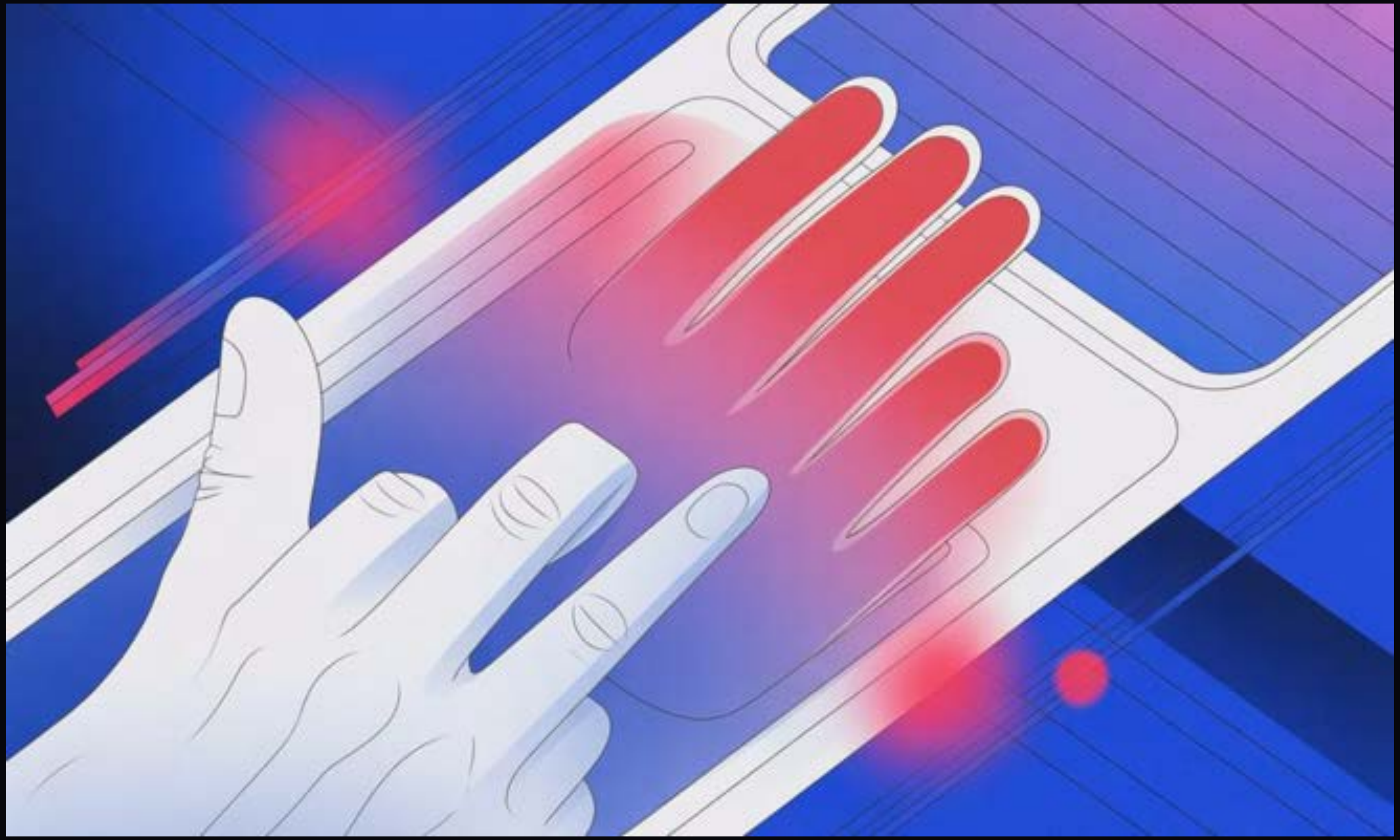
- Learning Rate Reduction on Plateau: factor=0.5, patience=4
- Early Stopping: patience=8, preventing overfitting
- Training Duration: **~40 epochs** over 35 minutes
- Best Model: Checkpoint saved based on validation accuracy

EXPLAINABILITY

Grad-CAM: Visualizing Model Attention

Our Explainable AI (XAI) implementation uses Grad-CAM visualization. It computes gradient-weighted class activation maps from the final convolutional layer ('last_conv'), creating heatmaps that highlight spatial importance for classification decisions.

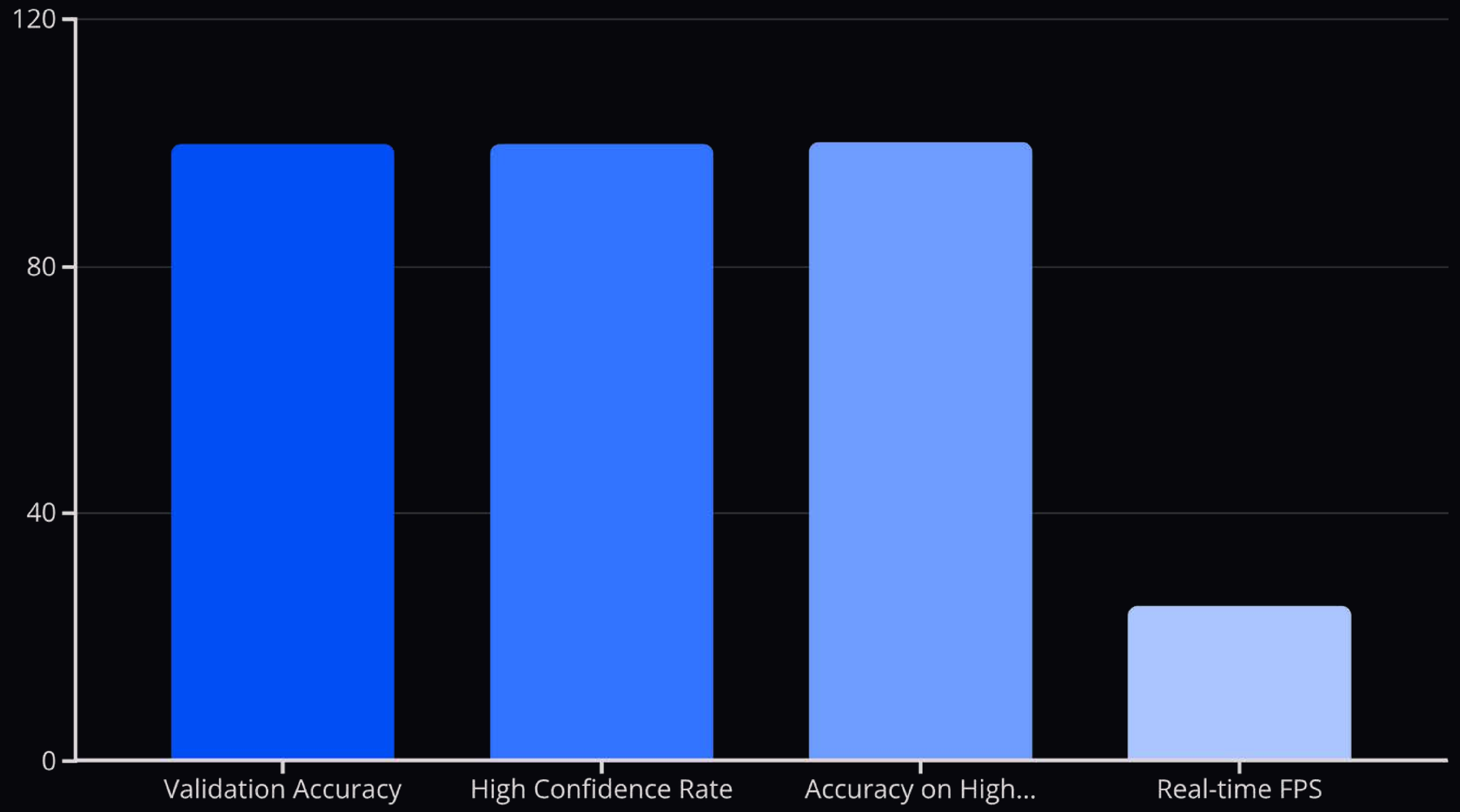
- Heatmap Generation: Multiplies feature maps by gradients, normalized to [0,1]
- Overlay: Resized and overlaid on original images using a jet colormap
- Interpretation: Red indicates high attention, blue indicates low attention



Consistently, Grad-CAM confirms the model focuses on semantically meaningful features like finger configurations and hand shapes, not background artifacts.

PERFORMANCE

Exceptional Accuracy & Confidence



The model achieved an outstanding **99.87% validation accuracy** across 3,200 test samples. Five gesture classes demonstrated perfect 100% accuracy, with the most challenging 'c-shape' still achieving 99.7%.

- Confusion Matrix: Virtually no inter-class confusion (only 4 total misclassifications)
- Confidence Distribution: 99.7% of predictions exceeded 85% confidence
- Low Confidence Predictions (<70%): **Only 6 samples**, with 50% accuracy, reliably indicating uncertainty.

Real-Time Deployment & Future Scope

Live Performance

Consistent 25-30 FPS with 30-40 ms inference latency on consumer hardware, suitable for interactive applications.

Resource Efficiency

30 MB model size enables deployment on laptops, embedded systems, and mobile platforms.

Future Enhancements

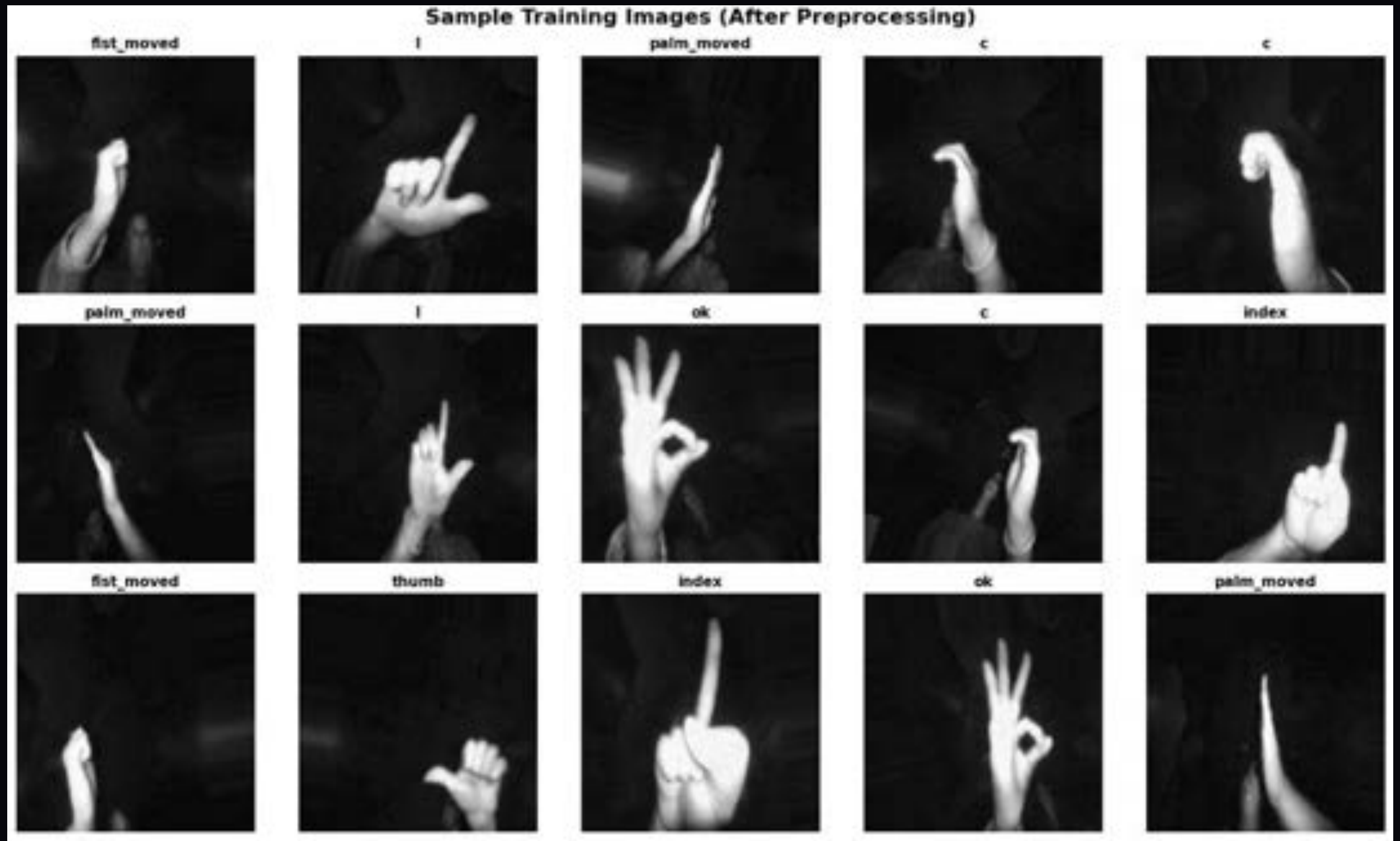
Dynamic gesture recognition with temporal models, multi-hand tracking, expanded gesture sets, and robustness testing across diverse conditions.

This work showcases practical machine learning engineering, making advanced computer vision accessible for research, assistive technologies, and human-computer interaction.

Input and Output Images

Output: Correct Predictions with High Confidence

Input Images after Preprocessing



Output: Incorrect Predictions - Error Analysis